

MA 102 Mathematics II
Ordinary Differential Equations
Lecture 18
(April 17, 2024)

Node

It is a critical point approached and also entered by each path as $t \rightarrow \infty$ (or as $t \rightarrow -\infty$). Here, there are four half-line paths AO, BO, CO and DO which together with the origin make up the lines AB and CD. All other paths resemble paths of parabolas and as each of these paths approaches O, its slope approaches that of the line AB. Refer to Figure 1 below.

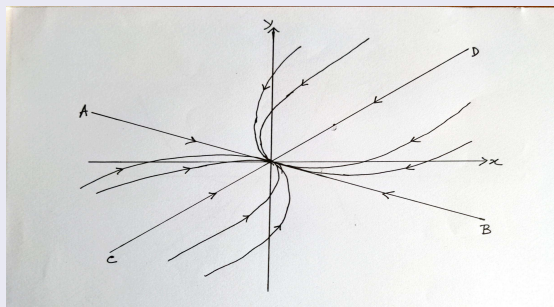


Figure 1

Example: Consider the system

$$\left. \begin{aligned} \frac{dx}{dt} &= x, \\ \frac{dy}{dt} &= -x + 2y. \end{aligned} \right\} \quad (1)$$

For this system, the origin is the only critical point.

Solution

Here, $a_1 = 1$, $b_1 = 0$, $a_2 = -1$, $b_2 = 2$ and hence, the auxiliary equation $m^2 - (a_1 + b_2)m + (a_1b_2 - a_2b_1) = 0$ for this system is $m^2 - 3m + 2 = 0$ giving us $m = 1, 2$.

The general solution can be found as

$$\left. \begin{aligned} x &= c_1 A_1 e^t + c_2 B_1 e^{2t} = c_1 e^t, \\ y &= c_1 A_2 e^t + c_2 B_2 e^{2t} = c_1 e^t + c_2 e^{2t}. \end{aligned} \right\} \quad (2)$$

When $c_1 = 0$, then $x = 0$, $y = c_2 e^{2t}$. In this case, the path is the positive y -axis when $c_2 > 0$ and the negative y -axis when $c_2 < 0$. Each path approaches and enters the origin as $t \rightarrow -\infty$.



When $c_2 = 0$, then $x = c_1 e^t$, $y = c_1 e^t$. This path is the half-line $y = x$, $x > 0$ when $c_1 > 0$, and the half-line $y = x$, $x < 0$ when $c_1 < 0$. Again both paths approach and enter the origin as $t \rightarrow -\infty$.

When both c_1 and c_2 are not zero, the paths lie on the parabolas $y = x + (c_2/c_1^2)x^2$, which go through the origin with slope 1. It should be understood that each of these paths consists of only part of a parabola, the part with $x > 0$ if $c_1 > 0$, and the part with $x < 0$ if $c_1 < 0$. Each of these paths also approaches and enters the origin as $t \rightarrow -\infty$; this can be seen at once from (2).

We can also proceed directly from (1):

$$\frac{dy}{dx} = \frac{-x + 2y}{x}, \quad (3)$$

which gives the slope of the tangent to the path through (x, y) [provided $(x, y) \neq (0, 0)$]. Then, on solving (3), we find $y = x + cx^2$.

This procedure yields the curves on which the paths lie (except those on the y -axis), but gives no information about the manner in which the paths are traced out. It is clear that the critical point $(0, 0)$ is a node for (1) and **it is unstable**. Refer to Figure 2 for complete visualization.

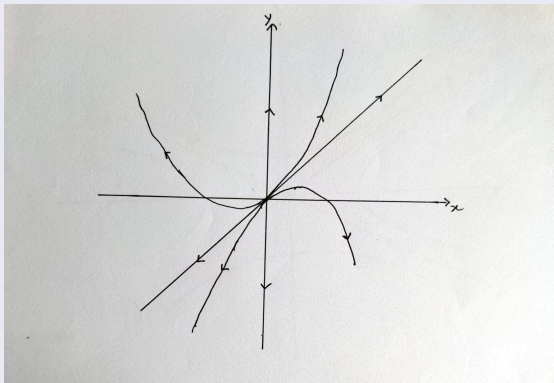
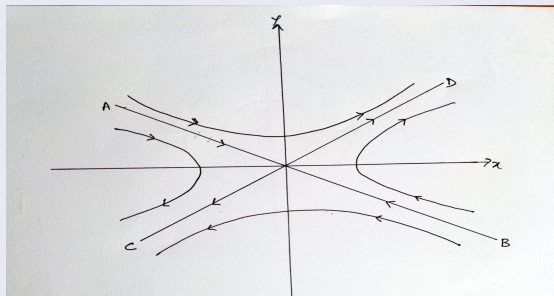


Figure 2

Saddle point

It is a critical point that is approached and entered by two half-line paths AO and BO as $t \rightarrow \infty$, and these paths lie on a line AB. It is also approached and entered by two half-line paths CO and DO (lying on line CD) as $t \rightarrow -\infty$.

Further to it, between the four half-line paths there are four regions – each containing a family of paths resembling (rectangular) hyperbolas. These paths do not approach O as $t \rightarrow \infty$ or as $t \rightarrow -\infty$ but are asymptotic to one or another of the half-line paths as $t \rightarrow \infty$ and as $t \rightarrow -\infty$. (the figure below)



Example

$$\frac{dx}{dt} = x, \quad \frac{dy}{dt} = -y.$$

Solution

Here, $a_1 = 1, a_2 = 0, b_1 = 0, b_2 = -1$. The auxiliary equation for the system is

$$m^2 - 1 = 0,$$

giving us $m = 1, -1$.

The general solution can be found as

$$\left. \begin{aligned} x &= c_1 e^t, \\ y &= c_2 e^{-t}. \end{aligned} \right\} \quad (4)$$



When $c_1 = 0$, then $x = 0, y = c_2 e^{-t}$. In this case, the path is the positive y -axis when $c_2 > 0$ and the negative y -axis when $c_2 < 0$. Each path approaches and enters the origin as $t \rightarrow \infty$.

When $c_2 = 0$, then $x = c_1 e^t, y = 0$. In this case, the path is the positive x -axis when $c_1 > 0$ and the negative x -axis when $c_1 < 0$. Each path approaches and enters the origin as $t \rightarrow -\infty$.

When both c_1 and c_2 are not zero, the paths lie on the (rectangular) hyperbolas $xy = c_1 c_2$, which will never go through the origin.

Clearly $(0, 0)$ is a saddle point which is unstable.



Center

A **center**, which is also sometimes called a **vortex**, is a critical point that is surrounded by a family of closed curves. It is not approached by any path as $t \rightarrow \infty$ or as $t \rightarrow -\infty$.

Example

The system

$$\frac{dx}{dt} = -y, \quad \frac{dy}{dt} = x \quad (5)$$

has the origin as its only critical point.

Here, $a_1 = 0, a_2 = 1, b_1 = -1, b_2 = 0$. Hence, the auxiliary equation $m^2 + 1 = 0$ gives us the roots as $m = \pm i$.

Therefore, the general solution of (5) can be written as

$$x = -c_1 \sin t + c_2 \cos t, \quad y = c_1 \cos t + c_2 \sin t. \quad (6)$$

The solution of (6) satisfying the conditions $x(0) = 1$ and $y(0) = 0$ is clearly

$$x = \cos t, \quad y = \sin t, \quad (7)$$

and the solutions determined by $x(0) = 0$ and $y(0) = -1$ is

$$x = \sin t = \cos(t - \pi/2), \quad y = -\cos t = \sin(t - \pi/2). \quad (8)$$

These two different solutions (7) and (8) define the same path C (Figure 4) which is evidently the circle $x^2 + y^2 = 1$. Both (7) and (8) show that the path is traced out in the anti-clockwise direction.

If we eliminate t between the equations of the system, we get

$$\frac{dy}{dx} = -\frac{x}{y}$$

whose general solution $x^2 + y^2 = c^2$ yields all the paths (but without their directions). Obviously the critical point $(0, 0)$ of the given system (5) is a center.

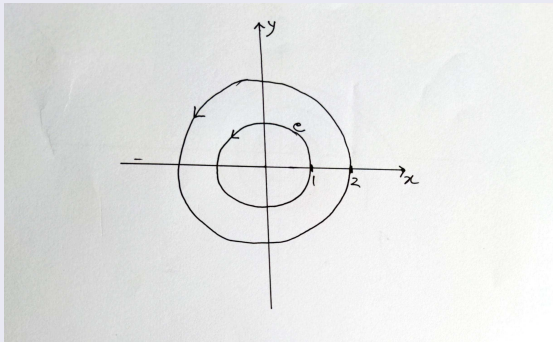


Figure 4

Spiral

A **spiral** or a **focus** is a critical point that is approached in a spiral-like manner by a family of paths that wind around it an infinite number of times as $t \rightarrow \infty$ or as $t \rightarrow -\infty$. Note that although the paths approach origin O , they actually do not enter it.

That is, a point P moving along such a path approaches O as $t \rightarrow \infty$ (or $t \rightarrow -\infty$) but the line OP does not approach any definite direction.

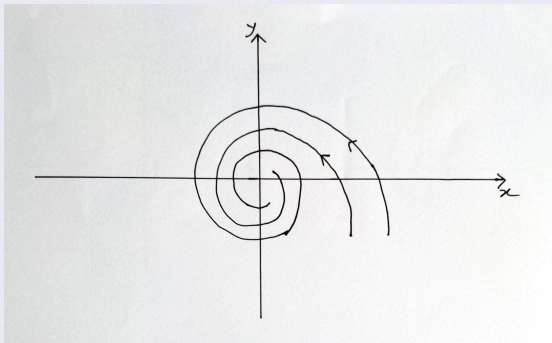


Figure 5

Example

If a is an arbitrary constant, then the system

$$\left. \begin{aligned} \frac{dx}{dt} &= ax - y, \\ \frac{dy}{dt} &= x + ay \end{aligned} \right\} \quad (9)$$

has the origin as its only critical point.

Solution

The differential equation of the paths

$$\frac{dy}{dx} = \frac{x + ay}{ax - y} \quad (10)$$

can be easily solved by putting $x = r \cos \theta$, $y = r \sin \theta$ from which we know $r^2 = x^2 + y^2$ and $\theta = \arctan(y/x)$.

Therefore, we have

$$r \frac{dr}{dx} = x + y \frac{dy}{dx} \quad \text{and} \quad r^2 \frac{d\theta}{dx} = x \frac{dy}{dx} - y.$$

With this result, (10) gives $\frac{dr}{d\theta} = ar$, which gives us

$$r = c e^{a\theta} \quad (11)$$

The two possible spiral configurations are for $a > 0$ and $a < 0$.

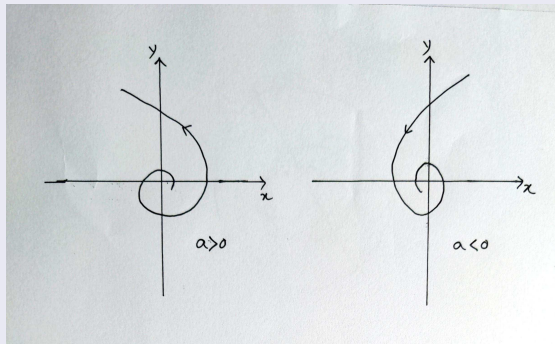


Figure 6



The directions in which these paths are traversed can be seen from the fact that $\frac{dx}{dt} = -ay$ when $x = 0$. If $a = 0$, then $r = c$ which is the polar equation of the family $x^2 + y^2 = c^2$ of all circles centered on origin.

In other words,

this example can be considered as the generalization of the previous example and since the center shown in Figure 4 stands on the borderline between the spirals of Figure 6, a critical point that is a center is often called a **borderline case**.